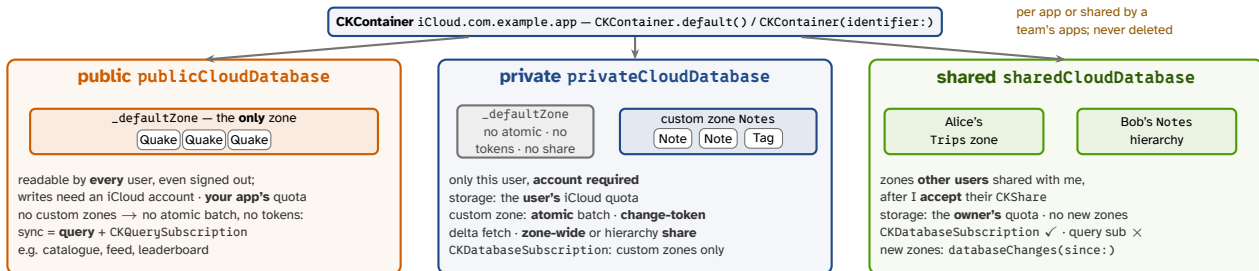


In one line: Apple's hosted database behind the user's iCloud account. A **container** holds three databases — **public** (everyone, your quota), **private** (this user, *their* quota), **shared** (zones others shared with me). **Records** live in **zones**; a save is checked against the record's **change tag**, so a conflict is an *error you resolve*, never a silent overwrite. Only **custom zones** give atomic writes, change tokens and sharing.



Records

- CKRecord = recordType + fields (String, number, Data, Date, CLLocation, arrays, CKAsset, Reference), ≤ **1 MB** without assets. CKRecord.ID = recordName (yours, e.g. UUID → idempotent retry) + zoneID: a record never changes zone.
- System fields: recordChangeTag, creationDate, modificationDate, creatorUserRecordID, parent, share. Local cache? keep encodeSystemFields(with:) or every save conflicts.
- CKRecord.Reference(recordID:action:): **same zone only**; .deleteSelf cascades. CKAsset(fileURL:): uploaded apart, fetched to a *temporary* file — copy it.

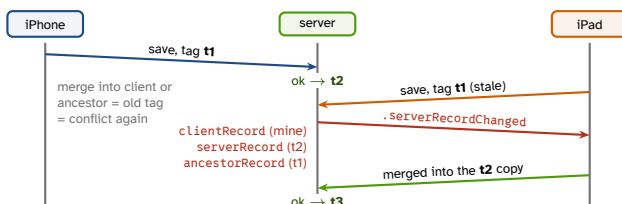
Operations vs async API

- Async** (iOS 15): record(for:), save(_:), modifyRecords(saving:deleting:...), records(matching:...), recordZoneChanges(...). Results are **per item** ([ID: Result]): no throw ≠ all saved.
- CKOperation** subclasses (CKModifyRecordsOperation, CKQueryOperation, ...): per-record callbacks, progress, cancel, dependencies, long-lived. Default QoS is low.
- Save policy: .ifServerRecordUnchanged (default, tag must match) · .changedKeys / .allKeys (overwrite, no check). isAtomic (default true): all-or-nothing *per custom zone*.

Example — save, and resolve a conflict

```
let db = CKContainer(identifier: "iCloud.com.example.app")
    .privateCloudDatabase
func rename(_ id: CKRecord.ID, to title: String) async throws {
    var rec = try await db.record(for: id) // carries change tag
    for _ in 1..3 {
        rec["title"] = title
        do { _ = try await db.save(rec); return } // tag must match
        catch let e as CKError where e.code == .serverRecordChanged {
            guard let server = e.serverRecord else { throw e }
            rec = server // merge INTO the server copy (new tag)
        }
        catch let e as CKError where e.retryAfterSeconds != nil {
            try await Task.sleep(for: .seconds(e.retryAfterSeconds ?? 1))
        }
    }
}
```

The change tag turns a race into an error



Queries, indexes, schema environments

- CKQuery(recordType:predicate:) + sort, paged by queryCursor. Indexes: **Queryable** (equality; on recordName to list all), **Sortable** (ranges, sort), **Searchable** (tokens).
- Development**: an unknown type or key is *created* on save. **Production**: it is an error. CloudKit Console → *Deploy Schema Changes*; production is **add-only**. Xcode runs use Development; **TestFlight + App Store use Production**.

Subscriptions + silent push

- CKQuerySubscription (public + private) · CKRecordZoneSubscription (one zone) · CKDatabaseSubscription (private + shared, every custom zone — how you hear of *new* shared zones).
- shouldSendContentAvailable = true → silent push (Background Mode *Remote notifications*, no prompt). **Coalesced, droppable**: a hint to fetch, not data.

Account

accountStatus(): .available · .noAccount · .restricted · .couldNotDetermine · .temporarilyUnavailable. On .CKAccountChanged, a new userRecordID() → **wipe old data**.

Errors you must handle — CKError.Code

serverRecordChanged	merge into e.serverRecord, save again
partialFailure	per item: e.partialErrorsByItemID
batchRequestFailed	atomic zone: another item failed; fix, resend
requestRateLimited	wait e.retryAfterSeconds, then retry — never a hot loop
zoneBusy	
serviceUnavailable	
limitExceeded	too many items / bytes: split the batch
changeTokenExpired	refetch the zone from a nil token
zoneNotFound	create the zone, retry
userDeletedZone	user wiped iCloud data: ask before re-upload
quotaExceeded	user's iCloud full: tell them, stop retrying
notAuthenticated	no account: check accountStatus()

Traps

- ! Default zone “to keep it simple” — no atomic batch, tokens or share; create a custom zone.
- ! Resaving your own copy after a conflict — old tag, fails forever.
- ! Fine in Development, empty in Production — schema not deployed.
- ! Private-DB quotaExceeded is the *user's* storage, not yours.

Remember

Container → **3 DBs** → **zones** → **records**. Custom zone = atomic + tokens + share; stale tag = merge.